Starlink for Texas school district
SpaceX is gearing up to make satellite Internet available to the education sector in Texas. **http://dgit.in/nov20-13**

Radiation in quantum computers
Cosmic rays and ionising radiation could soon be a problem for quantum computing. **http://dgit.in/nov20-14**

cation systems, and the radio transceivers in mobile phones and wireless networking devices. These circuits allowed for sophisticated, low-cost end-user terminals and gave rise to the communication systems within our wireless gadgets today.

In the late 90s, engineers understood that wireless technology was not just about making cellular calls but also about sending massive amounts of data over the air and the idea birthed Wi-Fi. In 1997, the Institute of Electrical and Electronics Engineers (IEEE) created a new standard for wireless internet service called the IEEE802.11 which was used to set up a wireless local area network or WLAN. After organising the basic frame for this, the mouthful of a name was shortened to Wi-Fi. This innovation changed the tides of the world of communication. Interestingly, Wi-Fi operates in the 'garbage bands' of the radio spectrum, the 2.4GHz UHF band and the 5GHz band. The technology gained a majority of its popularity under the 802.11b standard (operates on the 2.4GHz band), however, now the newer 802.11ac standard is more popular since it can handle higher data transfer rates. Cellular data connectivity also went from 2G to 3G to 4G, and now 5G, each generation offering higher speeds of data transmission.

IOT AND THE FUTURE

With Wi-Fi becoming more and more commonplace, a number of cutting-edge technologies emerged that offered a new form of communication. Only, this didn't involve helping humans communicate with each other, but instead, it allowed for gadgets to 'talk' to each other. We are, of course, referring to IoT or Internet of Things. The standards that would govern these connections, however, started showing up back in the late 90s. The first one that really took off and is still used widely today is Bluetooth.

In an unlikely partnership, Ericsson, Nokia, Intel, IBM and a few other researchers developed this new wireless standard that would let devices connect locally and wirelessly. The standard opened up an exciting arena for wireless accessories since there was no need for internet connectivity. Keyboards, headphones, desktops, laptops, mobile phones, they could all be connected via Bluetooth. The technology operates on the 2.4GHz band of the spectrum but it used up quite a bit of power. This led to the development of low power, close-range wireless standards such as Zigbee and Z-Wave in the 2000s. These protocols were widely used for home IoT such as connected lights, security cams, and smart locks.

Wi Fi™ | THREAD GROUP | Bluetooth® SMART | INSTEON | Z WAVE | ZigBee®

Popular home automation protocols

Image Credit: Electronic House

DID YOU KNOW?

The Bluetooth standard was funnily named after a medieval Scandinavian King, Harald "Blatand" Gormsson who ruled Denmark and Norway from the year 958 to 985. His nickname was Blatand which literally translates from Danish to Bluetooth. This was because the King was famously known for his dead tooth, which had a dark blue-grey shade. The name was chosen since King Harald was famous for "connecting" or uniting Scandinavia just as the people behind the name wanted to unite the PC and cellular industries with Bluetooth.

However, since the Wi-Fi hardware became more compact and started consuming lesser power, it was adopted in this space increasingly. Following this, new wireless communication protocols such as RFID (Radio-frequency identification) and NFC (near-field communication) also were introduced and used in numerous electronic devices. These wireless technologies are also very power-efficient.

Even though the advancements in wireless technologies is massive, there's more to come. Companies such as Facebook and SpaceX are experimenting with lasers to beam internet from satellites to the Earth's surface and this technology, which is still in its nascent stages, may replace electromagnetic waves for wireless communications as it can handle vast quantities of data transfer. Then there's Qi wireless charging, which is seen on multiple high-end and flagship smartphones, as well as computer peripherals and even truly wireless earphones. While it is pretty limiting for now, since you have to place your device accurately over a charging pad, some companies are experimenting heavily with wireless power. For example, in South Korea, tests are being carried out on electric buses that receive wireless power from cables placed underneath the road's surface. This is done by using Shaped Magnetic Field in Resonance (SMFIR) technology. Then there's the next level of wireless charging that is being tested by scientists at the University of Massachusetts Amherst which allow smartphones to charge continuously out of thin air! The scientists have developed a device called "AirGen" that uses a natural protein to create electricity from moisture in the air. It's needless to say that innovations in the field of wireless technology are nowhere near being finished.